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Master of Finance

Financial Econometrics – ECON1195

Return Characteristics and Market Efficiency of Coca-Cola and PepsiCo: Evidence from 2024

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Introduction

The global non-alcoholic beverage industry, valued at over USD 1.35 trillion in 2023 and projected to grow steadily through 2030, is dominated by two giants: The Coca-Cola Company (KO) and PepsiCo Inc. (PEP). Together, these firms account for more than 30% of global market share, yet they differ significantly in their product focus, brand positioning, and financial risk profiles. This makes them ideal candidates for comparative financial analysis, particularly in understanding how large multinational corporations respond to market dynamics and deliver returns to investors.

This report investigates the daily stock returns of Coca-Cola and PepsiCo in 2024 through three complementary lenses. First, descriptive statistics provide a snapshot of return behavior, highlighting differences in central tendency, volatility, and distributional shape. Second, formal tests of the Efficient Market Hypothesis (EMH) assess whether past return patterns offer predictive value or whether the stocks follow a random walk, consistent with market efficiency. Finally, the Fama–French Three-Factor Model is applied to measure systematic risk exposures market, size, and value factors while identifying whether either company generated abnormal returns beyond what standard models predict.

By combining statistical evidence with financial interpretation, the report not only contrasts the performance of Coca-Cola and PepsiCo but also situates them within the broader defensive-versus-cyclical spectrum of consumer staples. This dual focus—on both theory and business implications ensures that the analysis speaks to investors, academics, and industry professionals alike.

I. Descriptive Statistics of Coca Cola vs Pepsi

Coca-Cola (KO) and PepsiCo (PEP) dominate the global non-alcoholic beverages industry, together holding over 30% of market share (PepsiCo ~18%, Coca-Cola ~14%) (Market.us, 2024). Although both firms operate within the same sector, their stock return characteristics in 2024 highlight important differences in stability and risk profile.

Measure	KO	PEP
Mean	0.016	-0.051
Median	0.048	-0.029
Std. Dev.	0.80	1.04
Min / Max	-2.59 / 2.47	-4.03 / 3.93
Skewness	0.04	-0.03
Excess Kurtosis	0.66	2.27

Table 1: Daily Return Metrics (*Source: R output)*

Coca-Cola demonstrated steady upside momentum in 2024, with returns skewed toward the positive end of the distribution, supported by modest but consistent daily gains. In contrast, PepsiCo's performance leaned negative, reflecting a comparatively weaker trajectory across the year (RStudio calculations). The risk profiles also diverged: Coca-Cola's narrower and more stable return distribution pointed to resilience against daily market swings, while PepsiCo's wider fluctuations highlighted its more volatile nature.

Figure 1. Histograms of KO and PEP Daily Returns (RStudio 2024)

The shape of the return distributions reinforces this contrast. Coca-Cola exhibited smoother movements with limited extremes, whereas PepsiCo displayed fat-tailed characteristics, indicating a higher probability of sudden, large shocks (RStudio calculations). This was evident in Value-at-

Risk measures, where PepsiCo faced deeper potential losses under stress scenarios, underscoring its elevated downside risk relative to Coca-Cola.

From an investor perspective, the implications are clear. Coca-Cola delivered steadier, risk-adjusted performance, aligning with its status as a defensive stock underpinned by its core beverage franchise and strong global distribution network. PepsiCo, by comparison, represented a riskier tactical play, consistent with its broader diversification across beverages and snacks and its exposure to more cyclical consumption patterns. Together, the two companies illustrate the defensive versus cyclical spectrum within the global non-alcoholic beverage industry.

II. Tests of the Efficient Market Hypothesis

The Efficient Market Hypothesis (EMH) in its weak form states that asset prices fully incorporate historical information, meaning returns follow a random walk and cannot be predicted from past values (Fama, 1970). If true, investors cannot earn abnormal profits using past return data. To evaluate whether Coca-Cola (KO) and PepsiCo (PEP) daily returns in 2024 conform to this theory, we apply two tests: the Ljung–Box autocorrelation test and the Chow–Denning variance ratio test.

1) Ljung–Box Test (checking for autocorrelation)

Null hypothesis (H_0): Returns are independent of past values up to 10 days, which is what we expect under a random walk.

Alternative hypothesis (H_1): Returns are linked to past values, meaning they are predictable.

2) Chow–Denning Variance Ratio Test (checking for random walk over several days)

Null hypothesis (H_0): The variance of returns scales normally over 2, 5, and 10 days, which is consistent with a random walk.

Alternative hypothesis (H_1): At least one variance ratio is different, showing either trending behavior or mean reversion.

Test	KO Result	PEP Result	Decision (5% level)
Ljung–Box Q (lag = 10)	Q = 10.736, p = 0.378	Q = 6.237, p = 0.795	Fail to reject H_0
Chow–Denning (k = 2, 5, 10)	CD1 = 1.095, CD2 = 1.070	CD1 = 1.037, CD2 = 0.906	Fail to reject H_0

Table 2. EMH Test Results for KO and PEP (Source: RStudio calculations).

The p-values for the Ljung–Box test is comfortably above 0.05 for both KO and PEP, and the Chow–Denning statistics are far below critical values at all levels.

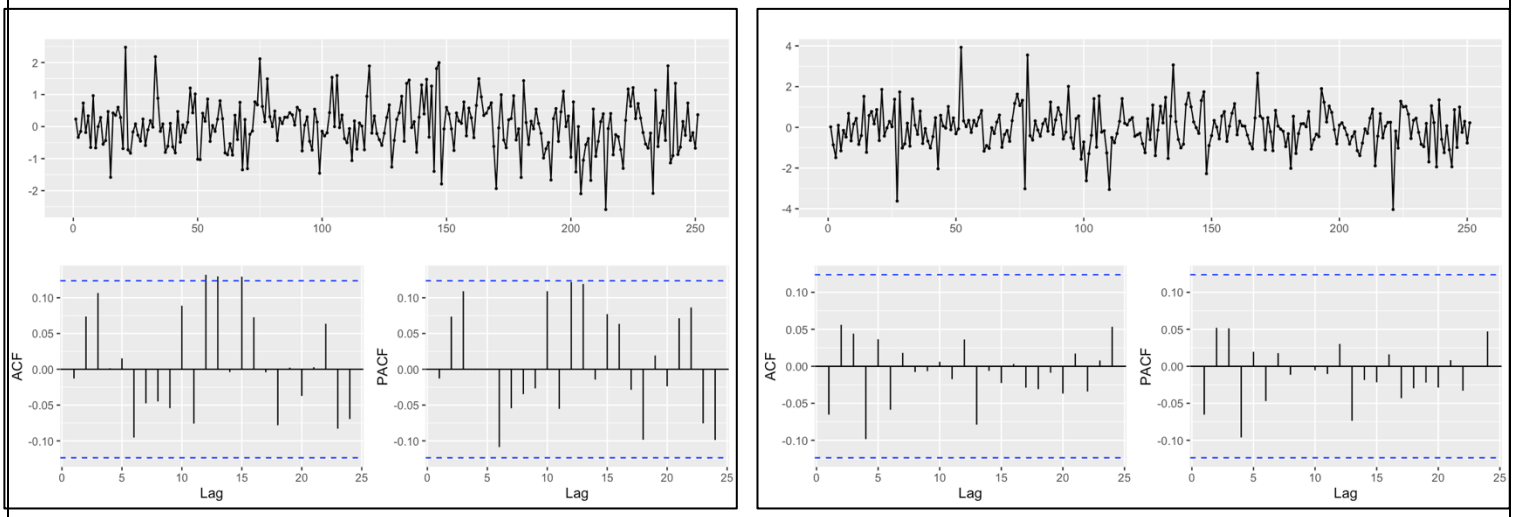

Figure 2: Correlogram of KO Daily Returns (left) and PEP Daily Returns (right)

The Ljung–Box test results (KO: p = 0.378; PEP: p = 0.795) indicate that we **do not reject H_0**. This means there is no evidence of autocorrelation up to 10 lags, and past daily returns do not predict future returns. Similarly, the variance ratio results show no significant departures from 1, and the Chow–Denning test statistics for both KO (1.095, 1.070) and PEP (1.037, 0.906) are well below the 10%, 5%, and 1% thresholds. We therefore again **do not reject H_0**, confirming that returns are neither trending nor mean-reverting over multi-day horizons.

In business terms, these findings suggest that both Coca-Cola and PepsiCo's stock returns in 2024 followed patterns consistent with a **random walk**. Investors could not exploit past price movements to generate excess profits. Despite PEP's higher volatility and fatter tails (from the descriptive statistics section), this did not translate into predictable patterns. The results are consistent with the **weak-form EMH**, showing that both KO and PEP operate in highly efficient, liquid markets where new information is quickly priced in.

Across both statistical frameworks, the evidence **supports** the **weak-form Efficient Market Hypothesis** for Coca-Cola and PepsiCo in 2024. Their daily returns show no significant serial correlation or predictable variance behavior. This means that trading strategies based on past return patterns are unlikely to provide consistent gains.

III. Fama–French Three-Factor Model Analysis

Overview

While the Capital Asset Pricing Model (CAPM) focuses solely on the market factor, empirical evidence suggests that other systematic risk factors also explain stock returns. The **Fama–French Three-Factor Model (FF3)** extends CAPM by adding size (SMB: small minus big) and value (HML: high book-to-market minus low) factors (Fama & French, 1993). The model helps to assess whether a company's returns are influenced by its exposure to the broader market, company size, or value versus growth orientation.

The model is specified as:

$$\boldsymbol{R_{\{i,t\}} - R_{\{f,t\}} = \alpha_i + \beta_M\left(R_{\{M,t\}} - R_{\{f,t\}}\right) + \beta_S SMB_t + \beta_H HML_t + \epsilon_t}$$

Where $\boldsymbol{R_{\{i,t\}} - R_{\{f,t\}}}$ is the excess return of stock iii over the risk-free rate, $\boldsymbol{\alpha_i}$ is the intercept (abnormal return), $\boldsymbol{\beta_M, \beta_S, \beta_H}$ are sensitivities to market, size, and value factors, and $\boldsymbol{\epsilon_t}$ is the error term.

We apply this model to Coca-Cola (KO) and PepsiCo (PEP) using 2024 daily return data, merged with the Fama–French daily factor dataset.

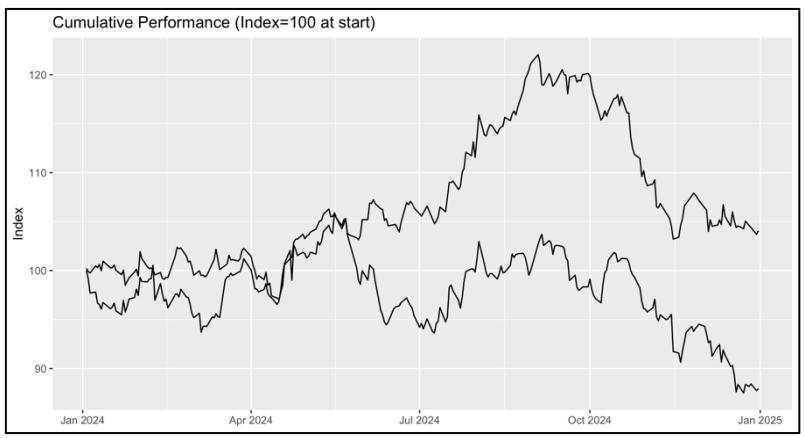

Figure 3: Cumulative Performance

KO Regression Output (RStudio results):

$$\boldsymbol{R_{\{i,t\}} - R_{\{f,t\}} = \alpha_i + \beta_M\left(R_{\{M,t\}} - R_{\{f,t\}}\right) + \beta_S SMB_t + \beta_H HML_t + \epsilon_t}$$

$$\boldsymbol{= -0.0064 + 0.0746 \cdot (MktRF_t) - 0.0234 \cdot SMB_t + 0.1345 \cdot HML_t + \epsilon_t}$$

PEP Regression Output (RStudio results):

$$R_{\{i,t\}} - R_{\{f,t\}} = \alpha_i + \beta_M\left(R_{\{M,t\}} - R_{\{f,t\}}\right) + \beta_S SMB_t + \beta_H HML_t + \epsilon_t$$

$$= -0.0759 + 0.0992 \backslash cdot\ (MktRF_t) - 0.1151 \backslash cdot\ SMB_t + 0.2079 \backslash cdot\ HML_t + \backslash epsilon_t$$

Variable	KO Coeff. (p-value)	PEP Coeff. (p-value)	Interpretation
Intercept (α)	–0.006 (0.90)	–0.076 (0.25)	Both alphas are insignificant → no abnormal returns.
Market β	0.075 (0.29)	0.099 (0.27)	Weak exposure to market excess returns.
SMB (size factor)	–0.023 (0.75)	–0.115 (0.22)	Negative but not significant → both behave like large-cap stocks.
HML (value factor)	0.134 (0.07)	0.208 (0.03)	KO shows mild value tilt; PEP shows significant value exposure.
R^2	0.015	0.021	Very low explanatory power at daily frequency.

Table 3: Fama–French Three-Factor Regression Results (2024)

The **alpha coefficients** for both companies are insignificant (KO: –0.006, p = 0.90; PEP: –0.076, p = 0.25). This means neither company produced abnormal daily returns once systematic factors were controlled, consistent with the **Efficient Market Hypothesis** (Fama, 1970).

Both stocks exhibit **low and insignificant market betas** (KO: 0.075; PEP: 0.099). This reflects their role as **defensive consumer staples**, less sensitive to broad market movements than cyclical firms.

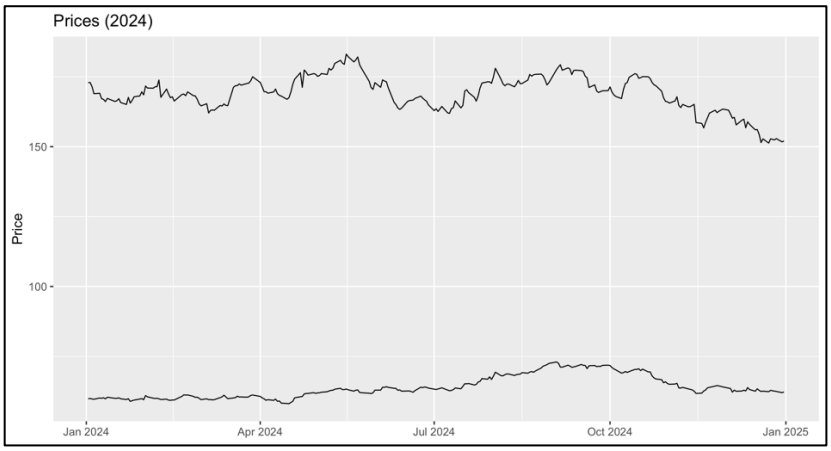

Figure 4: Price Trends of PEP (top) & KO (bottom) in 2024

For the **size factor (SMB)**, both KO (–0.023) and PEP (–0.115) show negative exposures, though statistically insignificant. This is expected since both are **large-capitalization multinationals**, and their returns do not move with small-cap risk premiums.

The most meaningful finding lies in the **value factor (HML)**. KO's coefficient is positive (0.134) and marginally significant at the 10% level, indicating some alignment with value stock characteristics. PEP's HML loading is stronger (0.208) and statistically significant at the 5% level, confirming that PepsiCo behaved more like a **value stock** in 2024. This may reflect market sentiment viewing PepsiCo's diversified portfolio (snacks + beverages) as undervalued relative to growth peers. Coca-Cola, in contrast, was priced closer to a stable, defensive growth firm, showing only weak value sensitivity.

However, the **R^2 values** are extremely low (KO = 0.015; PEP = 0.021), showing that the FF3 model explains only 1–2% of variation in daily returns. This is typical of high-frequency data, where short-term noise and firm-specific factors dominate (Malkiel, 2019). At monthly horizons, factor models generally have greater explanatory power.

Business Implications

From a portfolio perspective, these results suggest that **neither KO nor PEP offered abnormal returns** during 2024 investors could not systematically beat the market by holding them. Both remain **defensive mega-caps**, with low sensitivity to the market factor. Yet their factor exposures differ in meaningful ways. Coca-Cola is the steadier stock, with weak factor linkages and only marginal value orientation. PepsiCo, however, displayed stronger and statistically significant exposure to the value premium, making it more relevant for investors seeking value-tilted strategies.

In the broader non-alcoholic beverages industry valued at USD 1.35 trillion in 2023 and projected to reach USD 1.62 trillion by 2030 (Mordor Intelligence, 2024) these findings align with company positioning. Coca-Cola acts as the **defensive anchor** stock, while PepsiCo functions as the **higher-risk, value-tilted tactical play**.

The Fama–French three-factor analysis confirms that both Coca-Cola and PepsiCo's daily returns in 2024 were largely consistent with efficient market behavior. Neither stock produced abnormal returns, and both carried low market sensitivity. Yet PepsiCo showed a statistically significant **value tilt**, while Coca-Cola maintained its role as a **low-beta defensive staple**. For investors, KO represents a stable core holding, while PEP may appeal to value-oriented strategies seeking exposure to consumer staples with broader cyclical ties.

Conclusion

The findings of this report present a nuanced picture of Coca-Cola and PepsiCo's financial performance in 2024. Descriptive statistics revealed Coca-Cola's relatively stable and consistent returns, with lower volatility and narrower distributions, reinforcing its reputation as a resilient, defensive stock. PepsiCo, in contrast, displayed greater variability, fatter tails, and a higher likelihood of extreme shocks, underlining its riskier profile. Despite these differences, tests of the Efficient Market Hypothesis indicated that both firms' returns followed random walk behavior, meaning that past data could not be used to systematically predict future performance. This underscores the efficiency and liquidity of their trading environments, consistent with weak-form EMH.

The Fama–French Three-Factor analysis further deepened these insights. Neither company exhibited statistically significant abnormal returns once systematic factors were accounted for, confirming the absence of exploitable inefficiencies. Both firms behaved like large-cap stocks with low market betas, aligning with their classification as global consumer staples. However, their value exposures diverged: Coca-Cola showed only a marginal alignment with value characteristics, while PepsiCo demonstrated a significant tilt toward the value factor, positioning it more prominently within value-driven investment strategies.

Taken together, these results illustrate how two firms in the same sector can occupy distinct roles within investor portfolios. Coca-Cola stands out as a dependable, low-volatility holding, offering steady risk-adjusted returns and acting as a defensive hedge in uncertain markets. PepsiCo, by contrast, emerges as a more tactical opportunity, appealing to investors seeking exposure to value characteristics and cyclical consumer demand. Ultimately, the evidence demonstrates that while both companies operate within efficient markets, their unique factor sensitivities and return dynamics provide investors with differentiated opportunities for portfolio construction, balancing defensive stability with value-oriented risk-taking.

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